Please check that this question paper contains_ 09 questions and 02 printed pages within first ten minutes.

MORNING

[Total No. of Questions: 09] 19 DEC 2023 [Total No. of Pages: 02]

Uni. Roll No. ....

Program: B.Tech. (Batch 2018 onward)

Semester: 4th

Name of Subject: Computer Architecture and Microprocessors

Subject Code: PCIT-108

Paper ID: 16237

Detail of allowed codes/charts/tables etc. Nil

Time Allowed: 03 Hours **Max. Marks: 60**

NOTE:

1) Parts A and B are compulsory
2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
3) Any missing data may be assumed appropriately

Part – A **[Marks: 02 each]**

Q1.

a) Write the purpose of auxiliary carry flag.

b) Why does the DMA priority over CPU when both request memory transfer?

c) How is the stack top address calculated?

d) Where are Vector Processor used?

e) Outline the role of synchronization used in inter process communication.

f) Point out the specifications of High-End-High-Performance Processors.

Part – B **[Marks: 04 each]**

Q2. Elaborate the various types of interrupts in 8085.

Q3. Write a program to add a data byte located at offset 0500H in 2000H segment to another data byte available at 0600H in the same segment and store the result at 0700H in the same segment.

Q4. Discuss the different mapping techniques used in cache memories and their relative merits and demerits.

Q5. Explain the different types of addressing modes of 8085 with example of each.

P.T.O.

Q6. With a neat diagram, discuss the Interfacing 8051 to LCD.

Q7. Illustrate the concept of memory hierarchy; specify the meaning and applications of all types of memories.

Part – C **[Marks: 12 each]**

Q8. Write an 8085 assembly language program to search the largest data in the array. In addition to the program also write the algorithm and explanation of each step with meaning of each instruction.

OR

Write an 8085 assembly language program to generate Fibonacci series. In addition to the program, also write the algorithm and explanation of each step with meaning of each instruction.

Q9. Explain the architecture of 8085, specifying the purpose of main components with the help of labelled diagram.

OR

Write the meaning and explanation of following instructions a) LXI b) RLC c) JNC d) INX e) ADC f) DAA

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